

Measurement Formula Reference

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Rectangle: $A = l \times w$, $P = 2l + 2w$. Triangle: $A = \frac{1}{2}bh$. Circle: $A = \pi r^2$, $C = 2\pi r$. Rectangular prism: $V = l \times w \times h$. Cylinder: $V = \pi r^2 h$. Area uses units²; volume uses units³.

Round 1 — Retrieve It

1. Write the formula for the area of a rectangle.

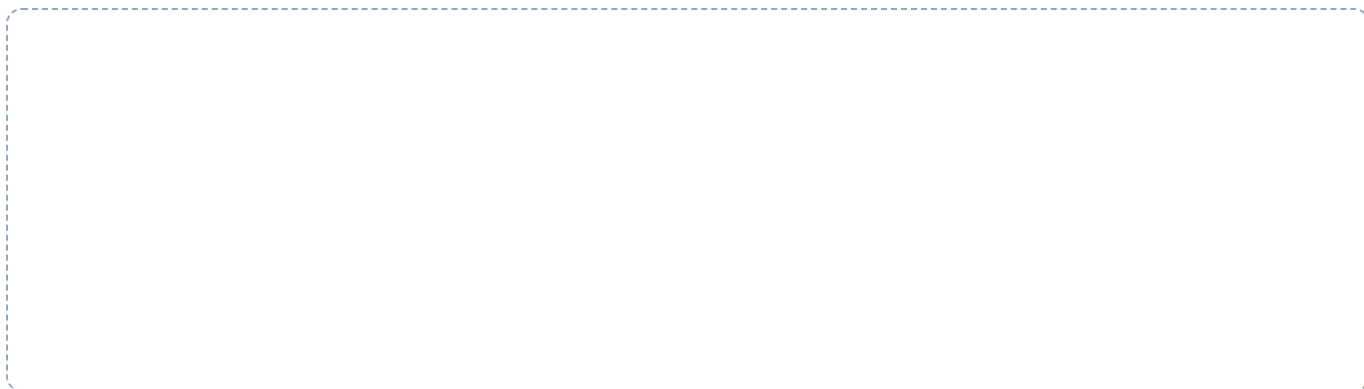
2. Write the formula for the area of a triangle.

3. Write the formula for the circumference of a circle.

4. Write the formula for the volume of a rectangular prism, and give the correct units label.

Round 2 — Sketch It

Sketch a rectangle and label its length and width. Beside it, write both the area formula and the perimeter formula, then circle the one that produces units².



Measurement Formula Reference — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A rectangle is 8 cm long and 3 cm wide. Find its area and its perimeter.

2. A triangle has a base of 10 in and a height of 6 in. Find its area.

3. A cylinder has a radius of 2 m and a height of 5 m. Find its volume. Use $\pi \approx 3.14$.

Round 4 — Reason from Evidence

A circle has a radius of 3 ft. Find both its area and its circumference. Explain why the area answer carries units² but the circumference answer does not. Use $\pi \approx 3.14$.

Measurement Formula Reference — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain how you substitute values into a formula and keep track of units. Use one example that is not printed on the cue card.

C

Connect

Connect the rectangle area formula $A = l \times w$ to the rectangular prism volume $V = l \times w \times h$. What does multiplying by the height add, and why does that change units^2 into units^3 ?

E

Extend

Create your own volume problem for a cylinder. Give the radius and height, then show the work using $V = \pi r^2 h$.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Measurement Formula Reference

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. $A = l \times w$.
2. $A = \frac{1}{2}bh$.
3. $C = 2\pi r$.
4. $V = l \times w \times h$, measured in units³.

Sketch It — look for

A labeled rectangle showing length and width, with $A = l \times w$ and $P = 2l + 2w$ written nearby. The area formula ($A = l \times w$) should be circled because it produces units².

Apply It

1. $A = l \times w = 8 \times 3 = \mathbf{24 \text{ cm}^2}$; $P = 2l + 2w = 2(8) + 2(3) = \mathbf{22 \text{ cm}}$.
2. $A = \frac{1}{2}bh = \frac{1}{2} \times 10 \times 6 = \mathbf{30 \text{ in}^2}$.
3. $V = \pi r^2 h = 3.14 \times 2^2 \times 5 = 3.14 \times 4 \times 5 = \mathbf{62.8 \text{ m}^3}$.

Reason from Evidence — look for

$A = \pi r^2 = 3.14 \times 3^2 = 3.14 \times 9 = 28.26 \text{ ft}^2$. $C = 2\pi r = 2 \times 3.14 \times 3 = 18.84 \text{ ft}$. Area multiplies two lengths, so the units multiply into units²; circumference is a single length around the circle, so it stays in plain units.

ACE — Extend look-for

Many answers possible. Check that the problem gives a radius and height and that the work correctly uses $V = \pi r^2 h$ with a units³ label.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.